KHALA RESEARCH · SECTOR REPORT

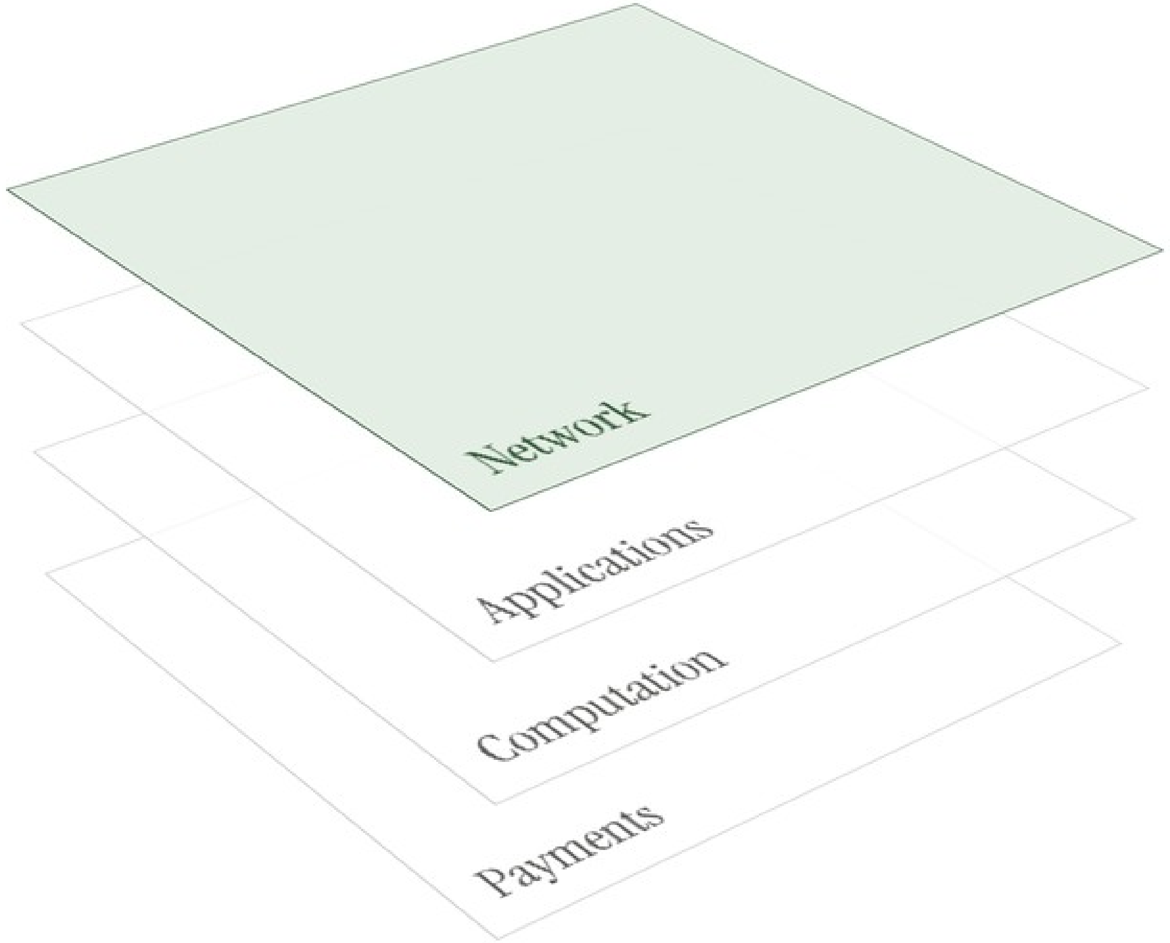
Privacy's missing layer

Open ledgers made crypto verifiable. Private networks will make it usable for business. Why confidentiality has become a condition of adoption, what investors have already paid for, and what it means for UR on Bittensor SN25.

Khala Research
Research team

November 2026
Data as of 15 October 2026

16 slides
Each one stands alone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EXECUTIVE SUMMARY

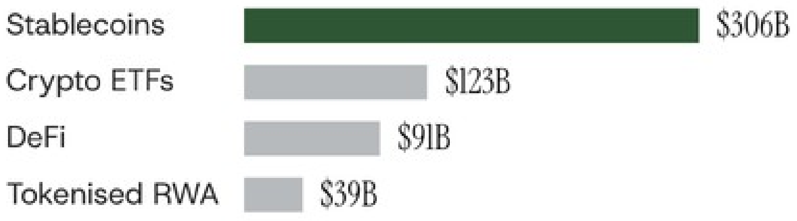
The privacy layer that hides who you are is unpriced, and UR is built for it

THE PROBLEM

$306B

Stablecoins on public ledgers

More than crypto ETFs, DeFi and tokenised assets combined. Every payment is visible.

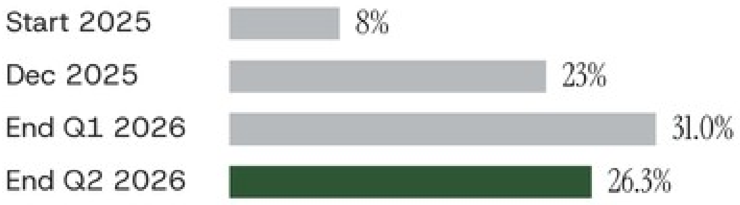


THE PROOF

$20.8B

Zcash market cap

A US Zcash ETF holds $463M, and the first confidential payroll and OTC trades ran in 2026.

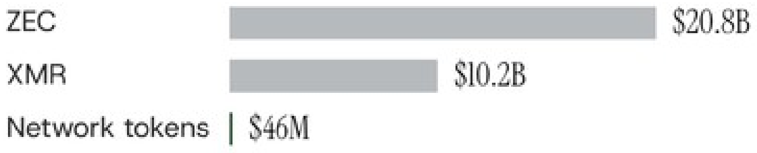


THE OPPORTUNITY

~$46M

All five network-privacy tokens combined

Encryption hides the payment but not the sender: in tests, a network observer linked over 95% of wallet users to their IP address from timing alone.



The bet

UR on Bittensor SN25

What it is

UR is the Bittensor subnet that pays people to route private traffic for URnetwork, whose VPN has about 350K monthly users (team-reported).

What would prove it

The proof is the first operator buying UR to deposit against real customer traffic, a link built into the design but set to zero for now.

What would break it

A conviction when Tornado Cash developer Roman Storm is retried in April 2027 would raise the legal risk for privacy developers, and the other threat is stakers pulling TAO from SN25 before deposits begin.

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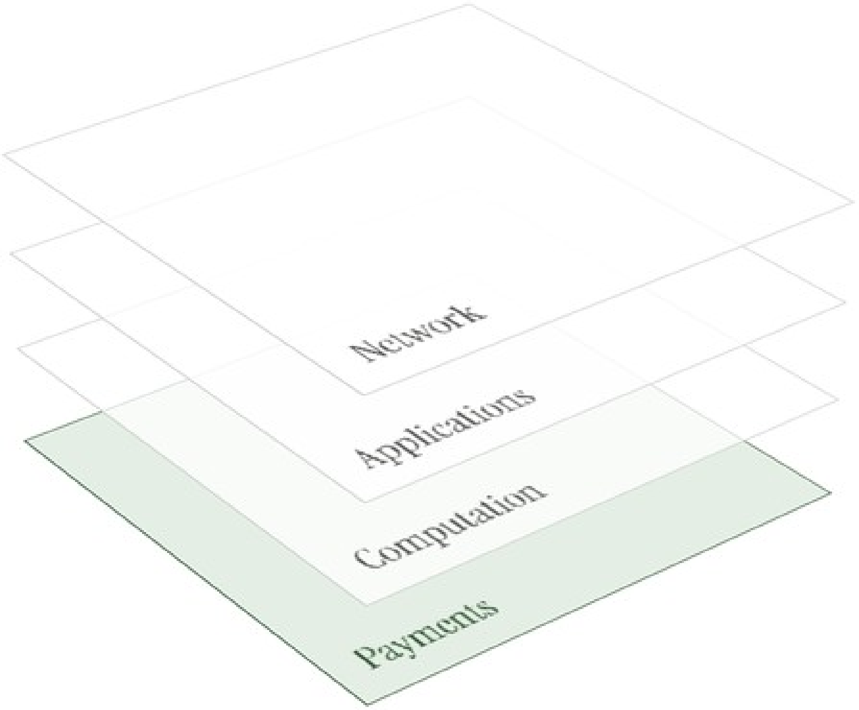
Why privacy, why now

Public ledgers now carry real business money, and AI makes them easy to read.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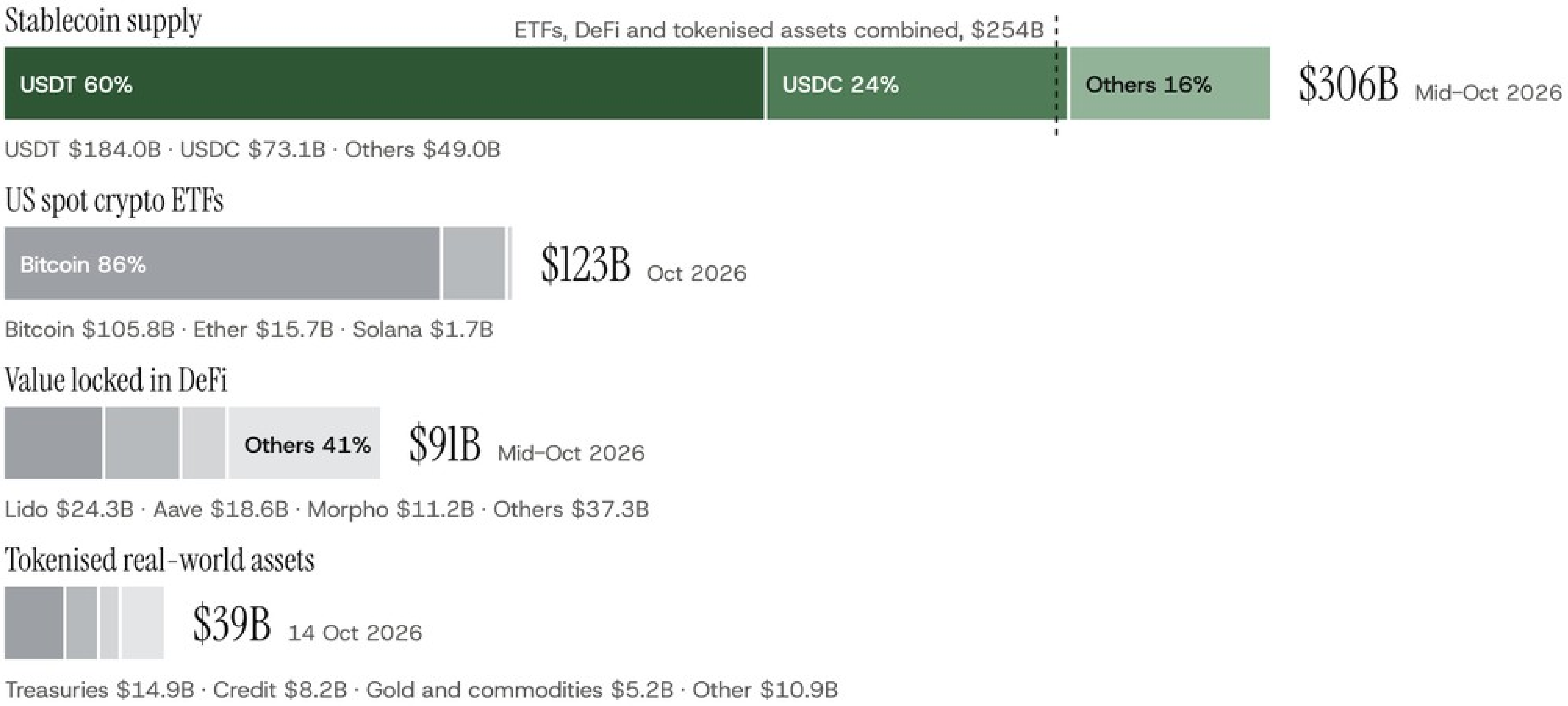
$306B of stablecoins sits on ledgers anyone can read

That is more than crypto ETFs, DeFi and tokenised assets combined, and every transfer is public.



Capital in selected onchain financial products, USD. ETF figures are net assets; tokenised assets are distributed value.

Anyone who links an address to a company can read its payroll, supplier terms and positions.

Tether and Circle issue 84% of the supply, so most stablecoin activity runs through two issuers' token contracts that anyone can watch.

At this scale confidentiality stops being a preference and becomes an operating requirement.

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Pay your team in stablecoins and your competitors can read the payroll

One linked address is all it takes. Here is one month at a fictional company.

Businesses don't need secrecy from auditors or regulators. They need to choose who reads their records.

With controlled disclosure the network still checks every payment, but only the company, its auditor and regulators see the details.

Payment	What the public ledger shows	What a competitor learns	What the public sees with controlled disclosure
Salaries	Three transfers totalling $29,450 on the 25th of each month	Headcount and individual pay	Three payments of [redacted] to [redacted] Verified
Supplier invoice	$184,000 to an address analytics tools label Kestrel Components	The main supplier, its price and terms	One payment of [redacted] to [redacted] Verified
Customer receipt	$420,000 in from an address labelled Orion Retail	A key customer and the contract size	One receipt of [redacted] from [redacted] Verified
Treasury transfer	$2.4M to an exchange deposit address	A likely sale, before it settles	One transfer of [redacted] to [redacted] Verified

Fictional company, names and amounts, for illustration.

FORCE ONE

Exposure gets cheaper

Models parse transaction histories at a scale that once took analysts weeks, so watching a public ledger costs close to nothing.

FORCE TWO

Private data gets more valuable

Useful assistants need email, records and preferences to work, so people and companies disclose more to them.

Demand for computation that uses data without exposing it

NEAR AI Cloud Live

Confidential inference since December 2025

Nillion Building

Multi-party computation, backed by Hack VC

Seismic Building

Secure hardware, backed by a16z

AI makes public ledgers easy to read and private data worth more

Both make it more expensive to operate in public.

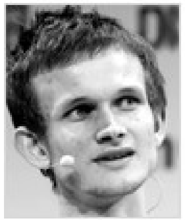
> “For users that prioritize privacy, this could become a ‘must-have’ feature.”

Zach Pandl, Head of Research, Grayscale, September 2026

His point: AI could push up demand for financial privacy.

Buterin and Durov both treat privacy as something you do not trade away

One runs a blockchain, the other a messaging app with over a billion users.



Vitalik Buterin

Co-founder

Ethereum

"Privacy is not about standing apart. It's about standing together."

Why I support privacy, 14 April 2025

Why it matters	Control over information is power, so privacy protects decentralisation, personal freedom and institutions such as the secret ballot.
Where it should live	In the protocol: payments, application activity, chain reads and network-level anonymity.
The gap in practice	A roadmap. Adoption depends on wallets and applications shipping it.

Same conclusion

Pavel Durov

Founder and CEO

Telegram, over 1B monthly users

"Telegram would rather exit a market than undermine encryption with backdoors and violate basic human rights."

Telegram channel, 21 April 2025

Why it matters	Encryption protects everyday communication and personal safety, and access built for authorities also opens a door for attackers.
Where it should live	In the product, with no backdoors, even at the cost of leaving a market.
The gap in practice	Telegram's default Cloud Chats are not end-to-end encrypted; only Secret Chats are.

Our read	Privacy has to be designed into wallets, chains and messaging from the start, which favours infrastructure over standalone privacy tools.

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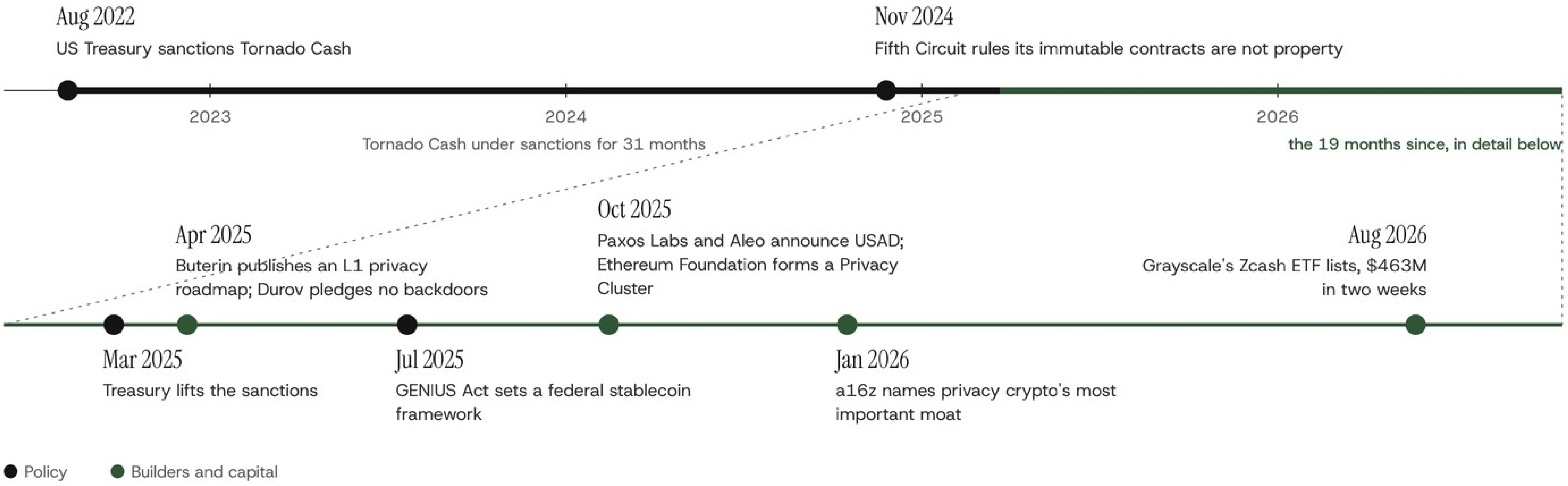
In four years the US went from sanctioning a privacy tool to listing a privacy ETF

Sanctions came off, builders moved in and investors followed, although the developer case is still open.



Aug 2022
US Treasury sanctions Tornado Cash

Nov 2024
Fifth Circuit rules its immutable contracts are not property

Tornado Cash under sanctions for 31 months

the 19 months since, in detail below

Mar 2025
Treasury lifts the sanctions

Apr 2025
Buterin publishes an L1 privacy roadmap; Durov pledges no backdoors

Jul 2025
GENIUS Act sets a federal stablecoin framework

Oct 2025
Paxos Labs and Aleo announce USAD; Ethereum Foundation forms a Privacy Cluster

Jan 2026
a16z names privacy crypto's most important moat

Aug 2026
Grayscale's Zcash ETF lists, $463M in two weeks

● Policy ● Builders and capital

Upper axis to scale from mid-2022; lower axis expands March 2025 to October 2026. Developer liability is still open: Roman Storm's retrial is set for April 2027.

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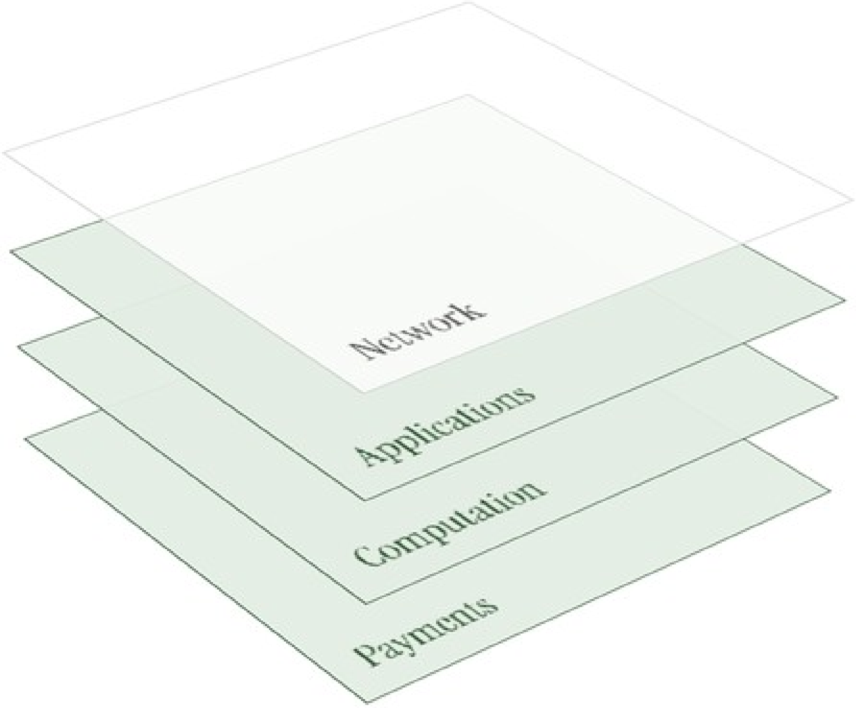
Who is paying for privacy

People already pay to hide what they send. Hiding who they are is still cheap.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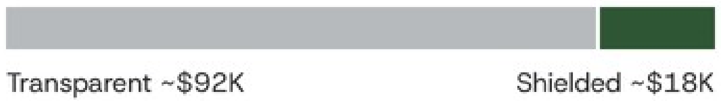
Zcash is worth $20.8B, but only a quarter of it is held privately

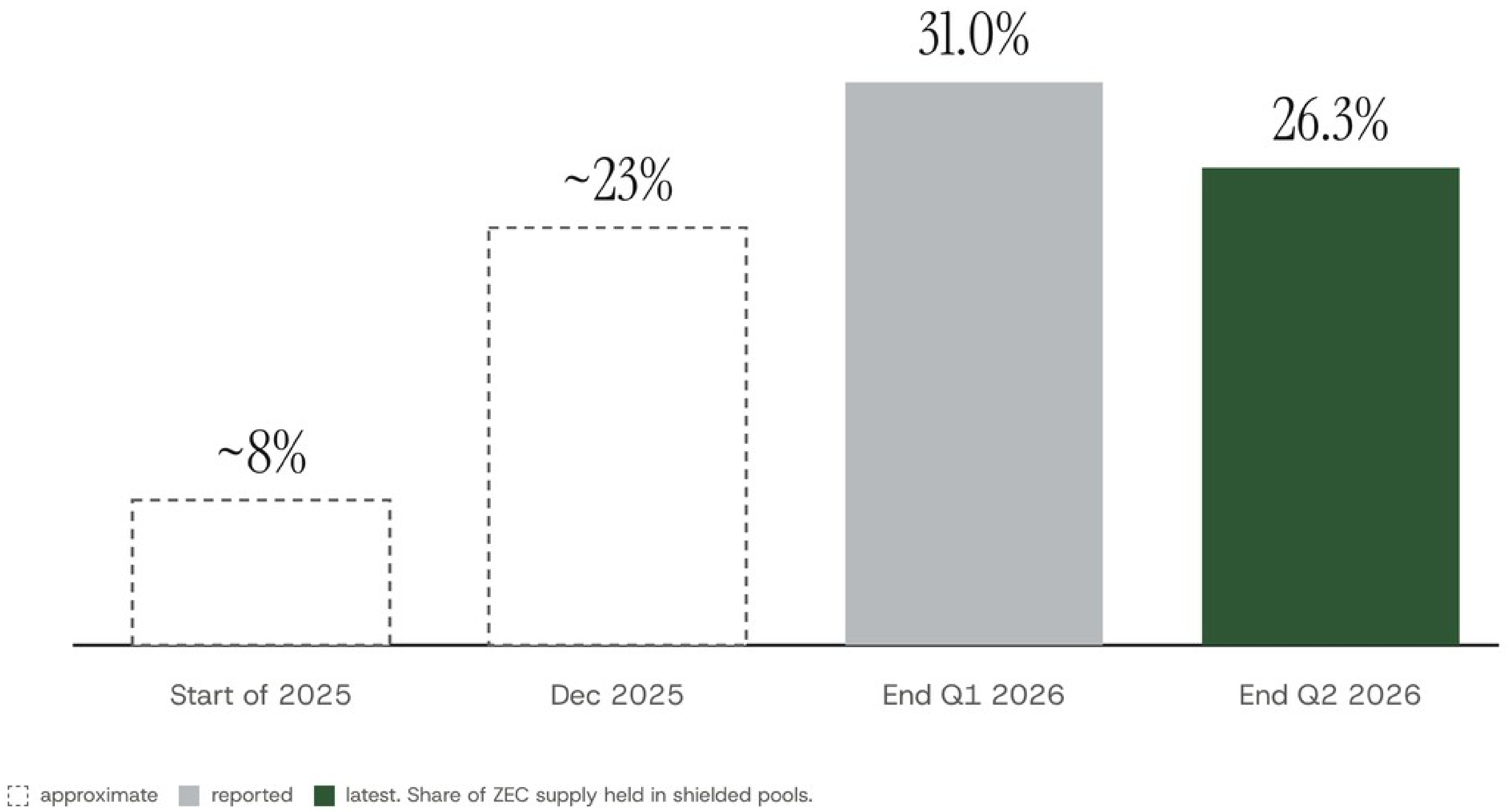
Shielded supply rose through 2025, then slipped back.

ZEC is worth about twice Monero and trades in a US-listed ETF, yet most activity still runs on the transparent side.

Investors are paying for the option of privacy ahead of its use.

Q2 2026 network fees





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Payroll and trading are going private first because leaks there cost real money

Hiding amounts is not enough on its own, because the network connection can still show who is transacting.

Use case	What must stay private	Evidence so far	Who is building or backing it
Payments and payroll *Company-reported deployments*	Salaries, supplier invoices and treasury movements	Bron paid salaries in confidential USDT on Ethereum mainnet through Zama, January 2026	Zama Bron Aleo
Trading and tokenised assets *Company-reported deployments*	Orders, positions and holdings	GSR executed a confidential OTC trade on the Zama protocol, March 2026; Dfns added confidential tokens to its wallet platform	Zama Dfns Pantera *investor*
Private computation and AI *NEAR AI live; others building*	Customer data and model inputs	NEAR AI Cloud runs confidential inference with an attestation per request; secure-hardware and MPC networks are building	NEAR AI Seismic a16z *investor* Nillion Hack VC *investor*
Programmable privacy *Building*	Hidden inputs for sealed-bid auctions, private ballots and strategy games	Reusable tools, so each application does not need its own cryptography	Fhenix Zama Multicoin *investor* Aztec Noir

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13 of the 18 privacy projects we track hide what you send. Only five hide who you are.

Each project sits in the layer it protects. In the network layer, UR and Nym are the two built so that no single server sees both ends.

Hides who you are 5 projects	Network Your IP and who you talk to	**UR (SN25)** Mesh routing	**Nym** Mixnet	**Orchid** Bandwidth market	**Sentinel** Decentralised VPN	**Mysterium** Decentralised VPN
Hides what you send 13 projects	Computation App and AI inputs	**NEAR AI** Secure hardware	**Seismic** Secure hardware	**Nillion** Multi-party compute	**Zama** Encrypted compute	**Nockchain** Early ZK proof of work
	Applications Contract state and activity	**Fhenix** Encrypted compute	**Aztec** ZK rollup	**Aleo** ZK layer 1		
	Payments and assets Balances and counterparties	**Zcash** Shielded pool	**Railgun** Shielded pool	**Privacy Pools** Shielded pool	**Umbra** Nimbus Solana wallet	**USAD** Private stablecoin

Khala Research classification by the main thing each project hides; several projects combine techniques.

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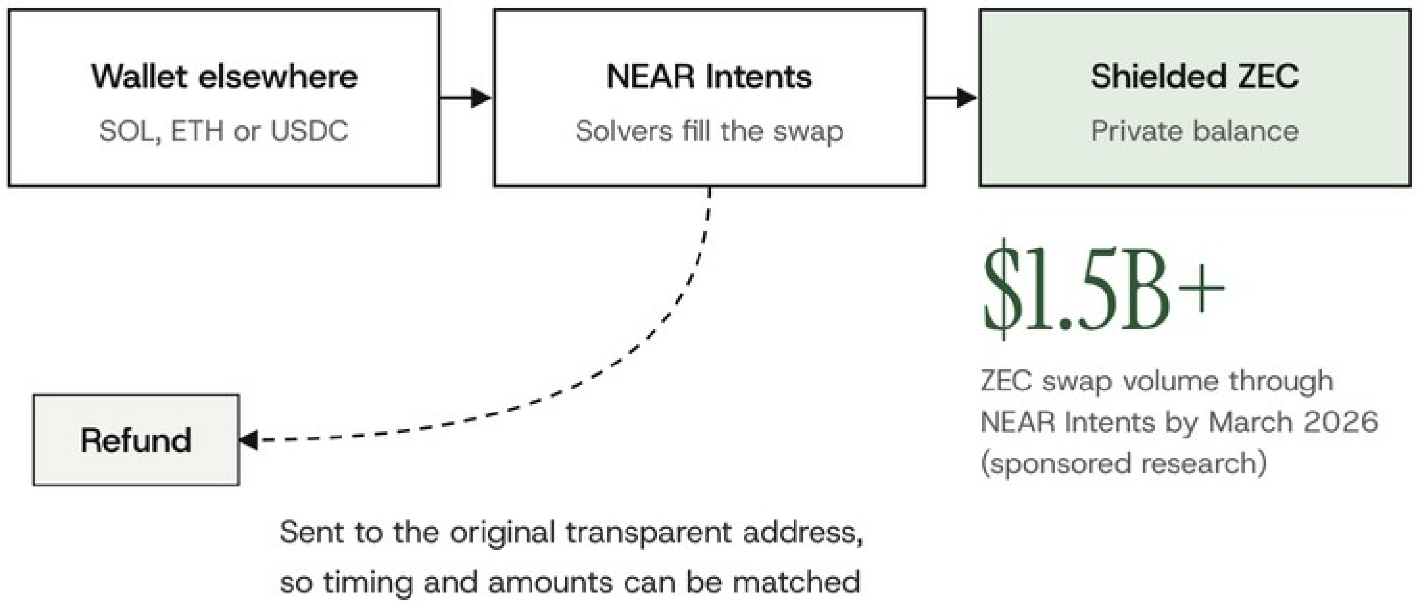
In tests, network timing alone tied over 95% of wallet users to their IP address

Privacy leaks where money crosses chains and where a wallet connects to the network.

Crossings leak: the swap route into private Zcash



Wallet elsewhere
SOL, ETH or USDC

NEAR Intents
Solvers fill the swap

Shielded ZEC
Private balance

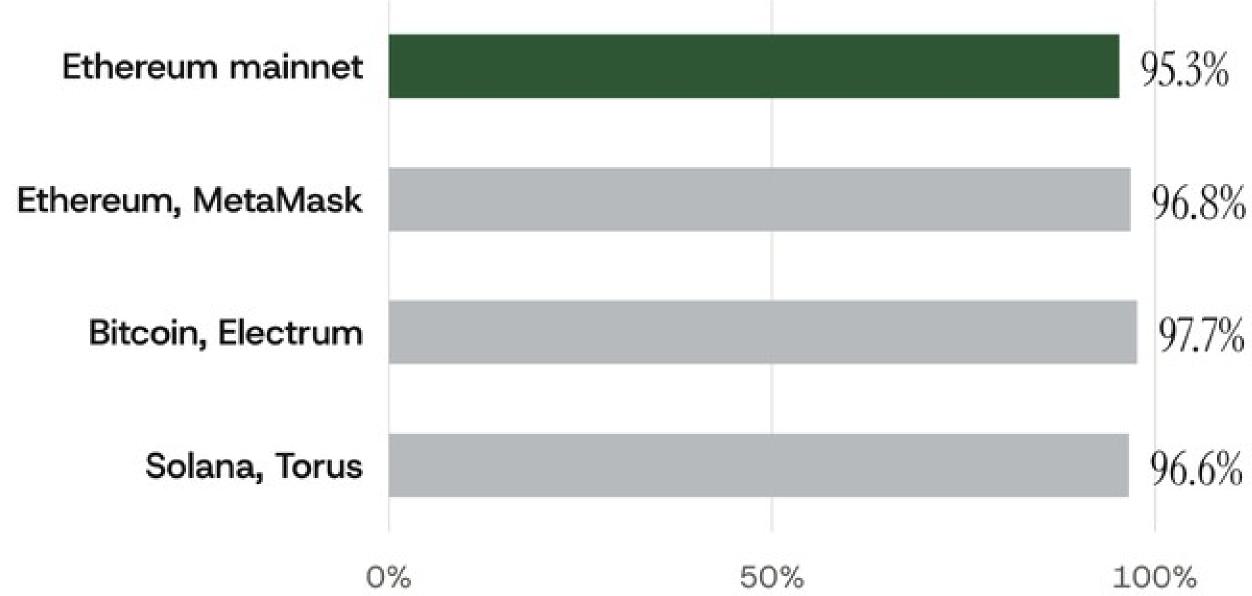
Refund

Sent to the original transparent address, so timing and amounts can be matched

$1.5B+

ZEC swap volume through NEAR Intents by March 2026 (sponsored research)

Connections leak: wallet users a passive observer identified



Wang et al., arXiv preprint, experiments March to April 2025. The observer watches traffic at internet exchanges or provider routers. Bitcoin and Solana rows ran on testnets. The authors name Tor or VPNs as possible defences.

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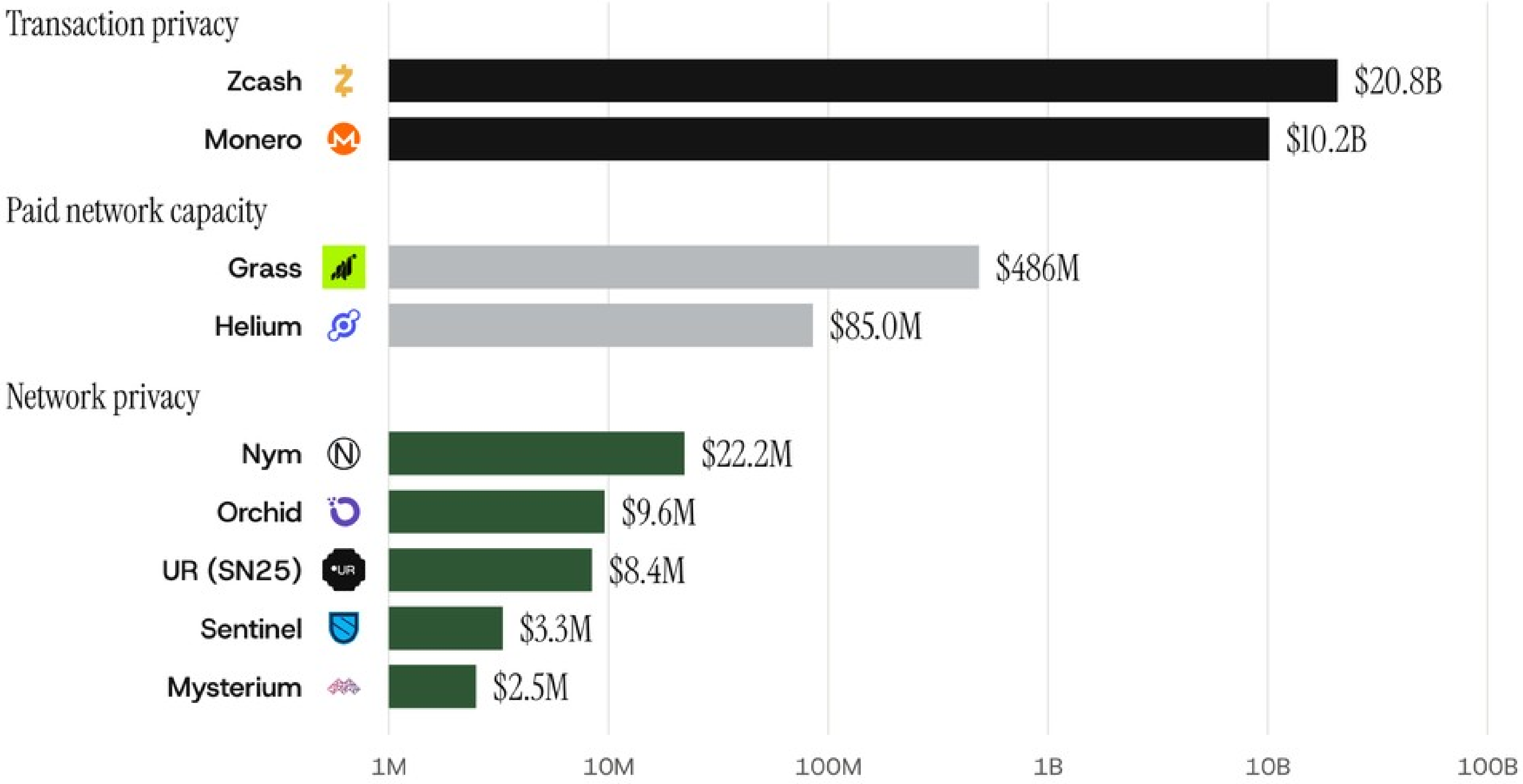
Hiding what you send is worth $31B. Hiding who you are is worth $46M.

A gap of about 670 times.

The five network-privacy tokens we track are worth about $46M together.

Grass and Helium, which outside customers pay for capacity, are worth far more.



Market cap, USD, log scale. CoinGecko and MEXC, 15 October 2026.

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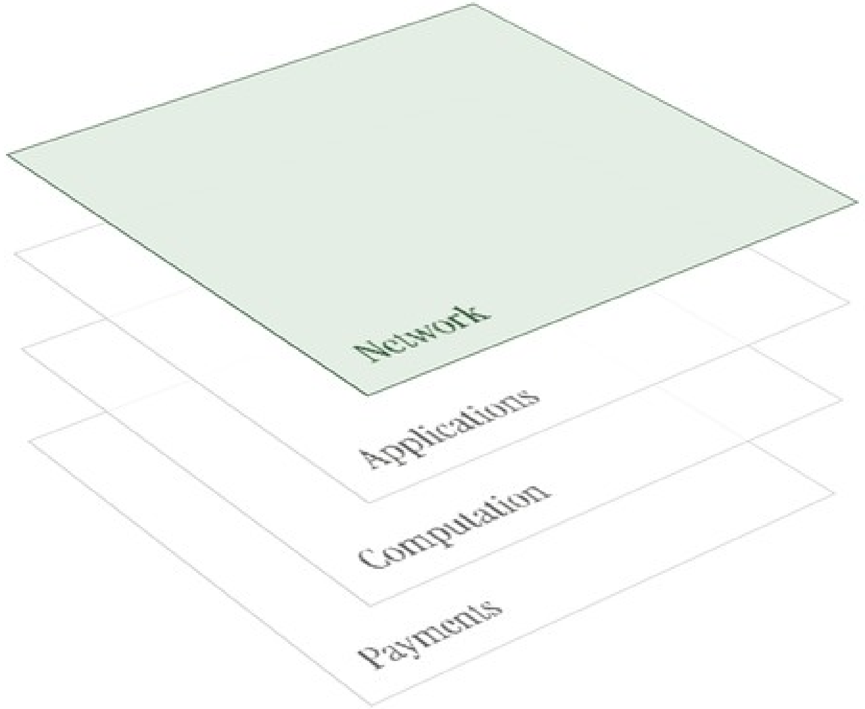
The case for UR

UR is one of very few ways to own the layer that hides who you are, and its product is already live.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Priced like Helium, UR is a 10x. Priced like Grass, it is a 58x.

URnetwork's VPN already has about 350K monthly users (team-reported), yet SN25 is valued at $8.4M, below both networks that customers already pay for capacity.

The gap is wide

Investors value Zcash and Monero at $31B, while the five network-privacy tokens we track are worth about $46M combined, a gap of roughly 670 times.

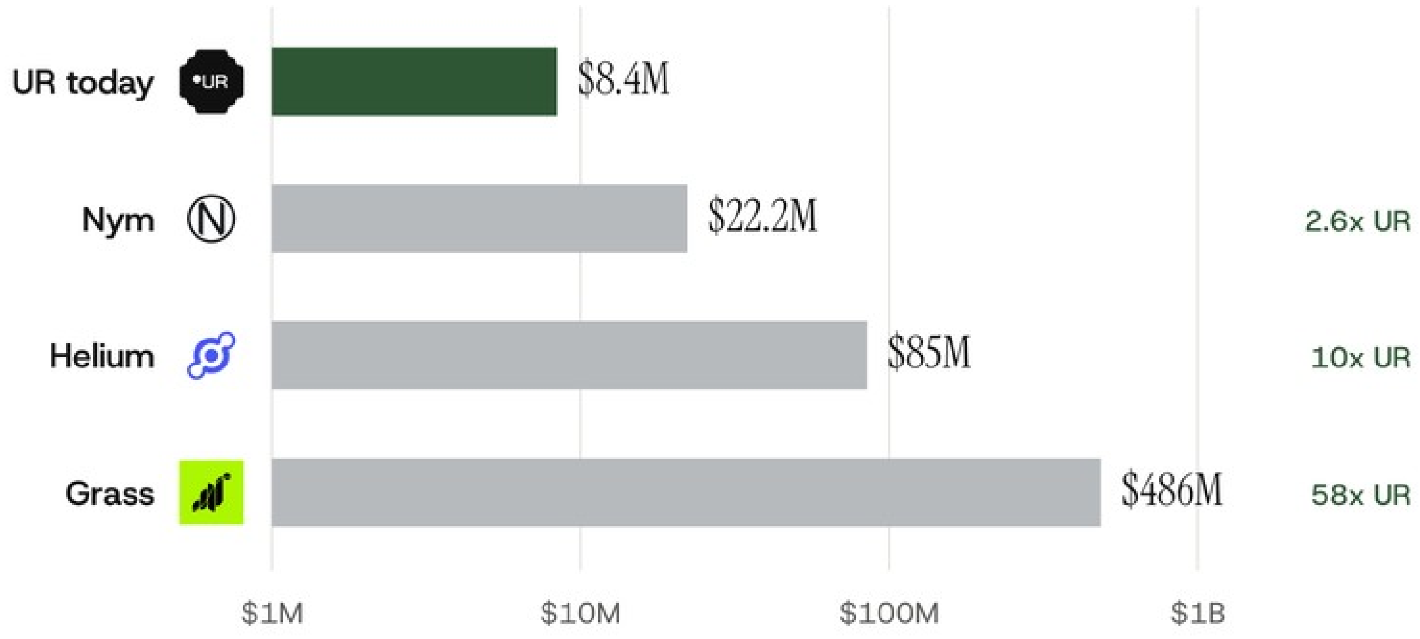
The product is live

URnetwork, the VPN that SN25 supplies, already has about 350K monthly users and 126,717 providers in 103 countries (team-reported), so UR is not waiting on a product.

The flywheel is built in

Once the deposit rate rises above zero, operators must buy and permanently lock UR in line with their traffic, so customer growth becomes buying pressure. Until then, TAO inflows pay for supply.

What UR would be worth at its peers' market caps



Market cap, USD, log scale. CoinGecko and MEXC, 15 October 2026. Multiples compare market caps today and are illustrative, not price targets.

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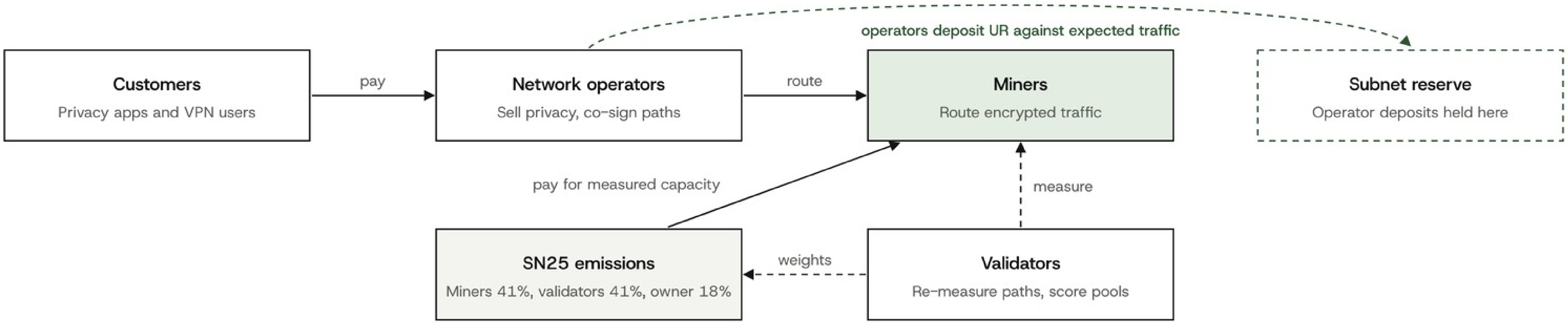
UR pays people to route private traffic, and Bittensor checks the work

Miners get paid only for capacity that validators have measured.



operators deposit UR against expected traffic

Customers
Privacy apps and VPN users

pay

Network operators
Sell privacy, co-sign paths

route

Miners
Route encrypted traffic

Subnet reserve
Operator deposits held here

pay for measured capacity

measure

SN25 emissions
Miners 41%, validators 41%, owner 18%

weights

Validators
Re-measure paths, score pools

Solid lines run today. Dashed lines are measurement, weights and the deposit link, which switches on once the reference rate is above zero. Deposits

126,717	Providers on URnetwork (team-reported)
103	Countries covered (team-reported)
~200	Top-level miner fleets, ranked by distinct exit IPs
$8.4M	SN25 market cap, mid-October 2026

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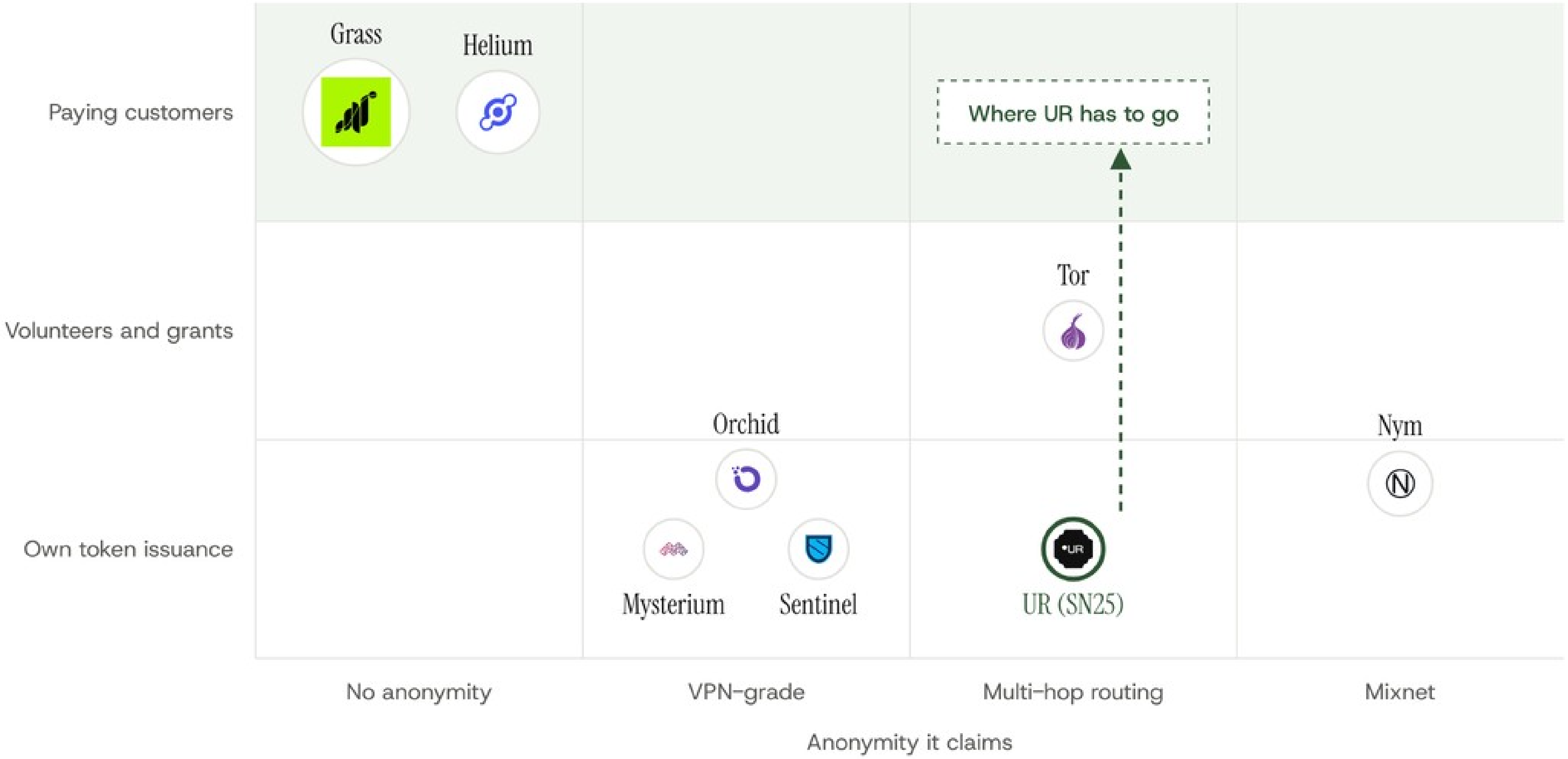
Nym claims stronger anonymity, so UR has to win on reach and price

UR's problem is who pays for its supply.

Nym has stronger anonymity and its own VPN. UR complements it rather than replacing it.

Token-paid networks stay at the bottom of this chart. Grass and Helium moved up by finding paying customers.



Circle size: market cap, CoinGecko and MEXC, 15 October 2026. Tor has no token.

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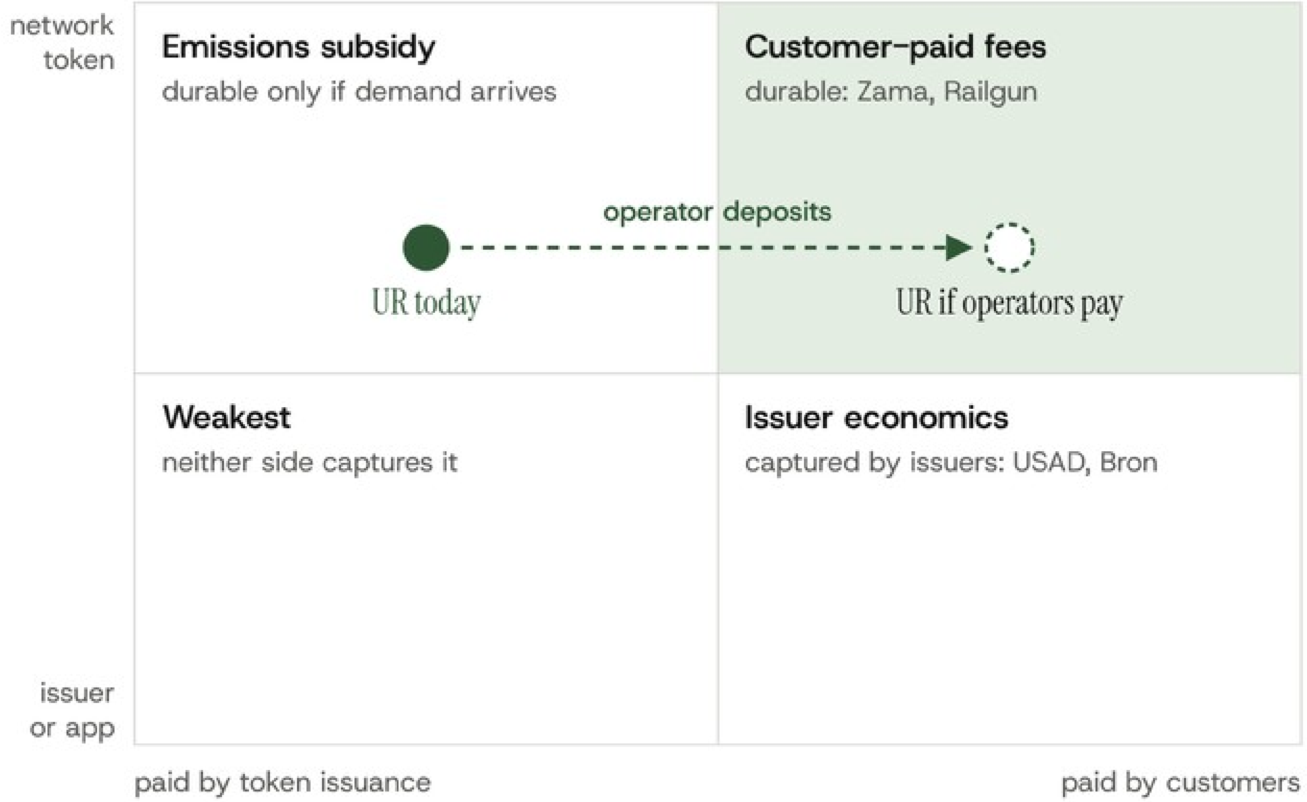
UR holders get no revenue today. The token runs on TAO inflows until operators pay.

The litepaper rules out any right to profits, and the operator deposit rate starts at zero.



Emissions subsidy
durable only if demand arrives

Customer-paid fees
durable: Zama, Railgun

Weakest
neither side captures it

Issuer economics
captured by issuers: USAD, Bron

How value reaches SN25, most likely first

1	TAO inflows	Bittensor sets each subnet's emissions by its net TAO inflows over about three months, so staking into SN25 lifts emissions and alpha demand.	Live since November 2025 Breaks if: net outflows cut emissions and miner pay together.
2	Sector re-rating	SN25 trades as Bittensor's privacy exposure while capital rotates into the category.	Live Breaks if: the most reversible pathway; a ZEC drawdown would likely drag it.
3	Operator deposits	Operators acquire UR to deposit against traffic, and deposits stay in the reserve rather than returning to circulation.	Rate set to zero at launch Breaks if: the rate stays at zero, or operators never arrive.
4	Reserve governance	Future onchain governance could direct how the reserve is used.	Speculative; no proposal Breaks if: the owner controls the contract until it hardens.

Ranked by likelihood. The deposit pathway is the one that turns UR from a subsidised network into a paid one.

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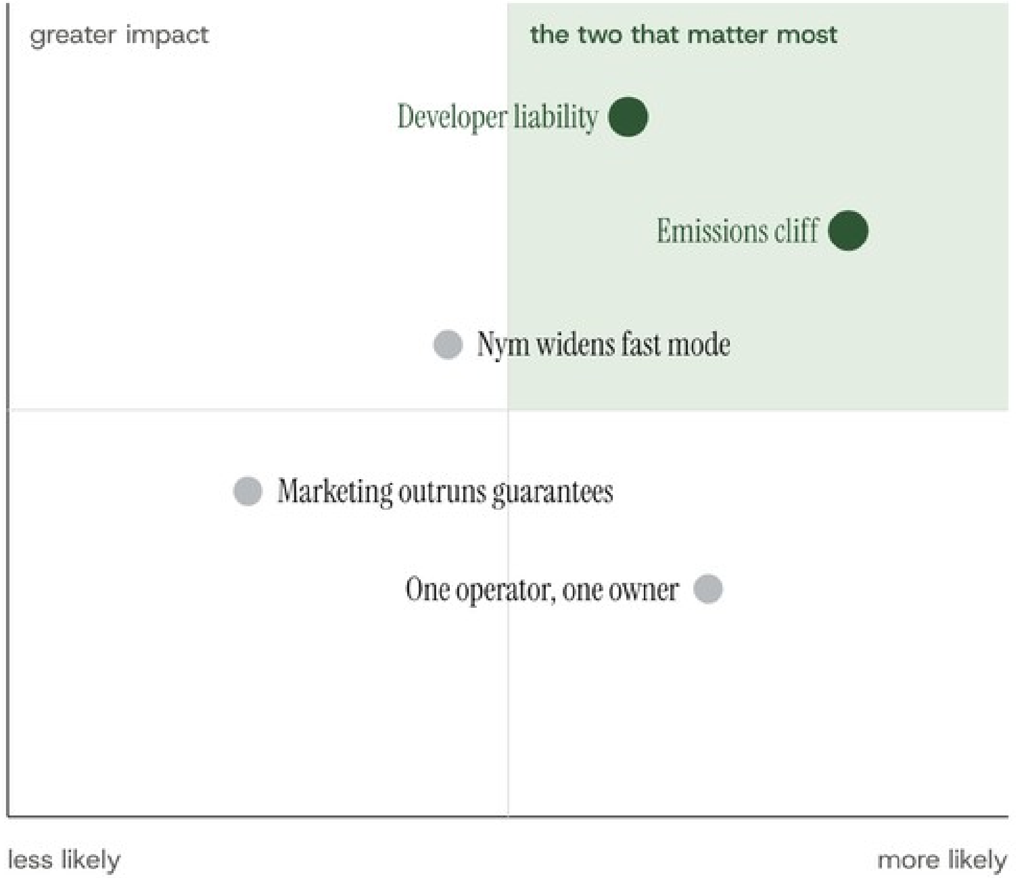
Two things could break UR: a court case and stakers walking away

The court rules in April 2027. Stakers decide every day.



Risk	What would have to happen	Signal to watch
Developer liability	Roman Storm's retrial on the laundering and sanctions counts, set for 26 April 2027, ends in a conviction and reopens exposure for privacy developers.	Rulings on the pending acquittal motion; the retrial verdict
Emissions cliff	Stakers pull TAO from SN25 before deposits begin, and miner pay and exit coverage shrink together.	Net TAO flow into SN25, month by month
Nym widens fast mode	Nym, with a VPN, a16z backing and rewards moving to real traffic, takes the operators UR needs.	Nym's browser and wallet integrations
Marketing outruns guarantees	URnetwork markets Tor-like anonymity while UR claims no defence against a global observer, and a public deanonymisation lands.	Independent audits of URnetwork's privacy claims
One operator, one owner	BringYour stays the only operator, the owner stays the largest validator, and its multisig keeps control of settlement with no upgrade timelock yet.	A second operator; the timelock going live

Khala Research judgement of likelihood and impact.

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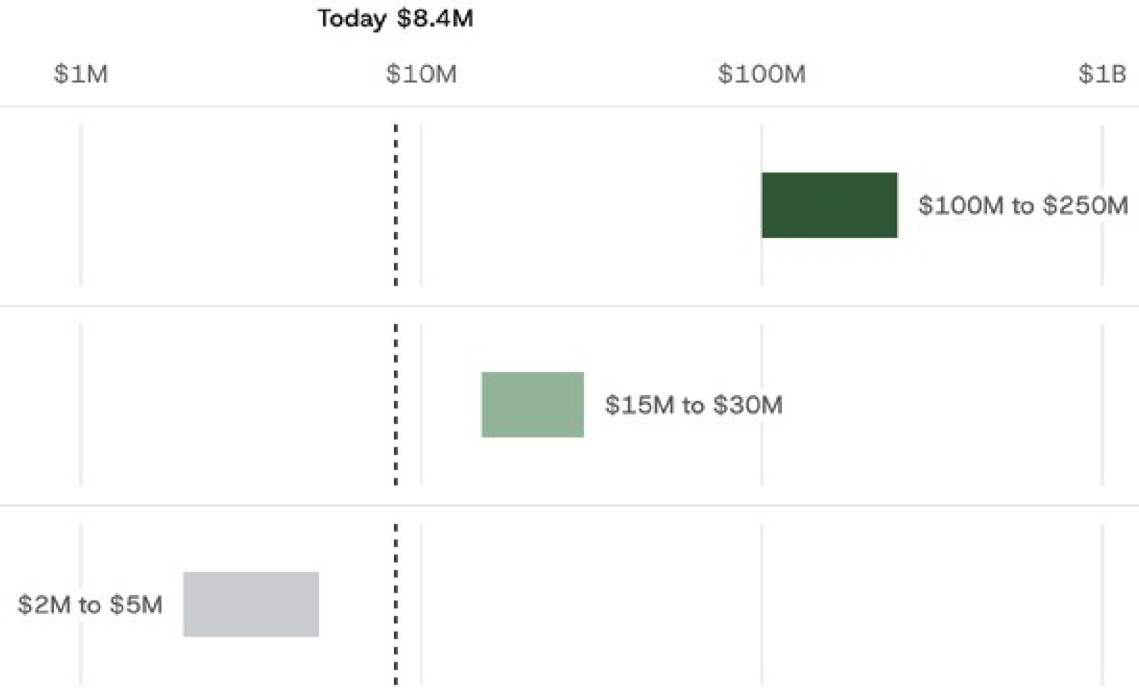
Our base case is 1.8 to 3.6x by the end of 2027, and the bull case is 12 to 30x

Which case plays out depends on one number: operator deposits on SN25, which are zero today.



Scenario	What has to happen	From today
Bull	Wallets and privacy apps buy capacity, and operator deposits reach a quarter of miner emissions.	12x to 30x
Base	Deposits switch on in 2027, and two to four operators pay small amounts.	1.8x to 3.6x
Bear	The deposit rate stays at zero, there is still one operator, and TAO flows out of SN25.	-76% to -40%

The number to watch

Monthly operator deposits on SN25. The first non-zero deposit moves UR out of the bear case, and deposits worth a quarter of miner emissions put it in the bull case.

Implied SN25 market cap by end 2027, log scale. Illustrative ranges, not forecasts or investment advice.

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Sources and notes

Market data is as of mid-October 2026 and will change. Verify live figures before acting on this analysis.

Deployments reported only by the companies involved are labelled. Scenario ranges are illustrative judgements, not forecasts.

This report is for information only and is not investment advice. Khala Research, its contributors and affiliated firms may hold positions in assets discussed here.

DefiLlama stablecoins and DeFi TVL, 15 Oct 2026

The Block US spot ETF net assets, 10 Oct 2026

rwa.xyz tokenised asset distributed value, 14 Oct 2026

CoinGecko, CoinMarketCap, MEXC, Taostats market data, 14 to 15 Oct 2026

UR litepaper and ur.xyz; URnetwork, ur.io (team-reported)

Vitalik Buterin Why I support privacy (14 Apr 2025); L1 privacy roadmap (11 Apr 2025)

Cointelegraph Durov on backdoors (21 Apr 2025); Zcash above $20B (7 Sep 2026)

TechCrunch Telegram passes 1B users (19 Mar 2025)

The Block Zcash shielded supply (26 Dec 2025); Ethereum Foundation Privacy Cluster (Oct 2025); Storm retrial (26 Aug 2026)

Blockworks Advisory Zcash token holder report Q2 2026, funded by GenZcash

CoinDesk Research Encryption Supremacy (31 Mar 2026), sponsored by GenZcash

Cryptopolitan ZachXBT on Zashi and NEAR Intents (21 Oct 2025)

NEAR AI NEAR AI Cloud and Private Chat (3 Dec 2025)

Zama; GSR; Aleo confidential payroll (16 Jan 2026); confidential OTC trade (12 Mar 2026); USAD (1 Oct 2025)

a16z crypto Privacy trends for 2026 (Jan 2026)

Willkie Farr Tornado Cash delisting (Mar 2025)

Wang et al. Time Tells All, arXiv 2508.21440 (preprint)

Nym Roadmap for 2026 (Mar 2026)

FalconX State of Bittensor (Taoflow emissions)

Bitrue Grass revenue, team-reported (30 Sep 2026)

crypto.news / Blockworks Helium data credit burn (Jun 2026)

CoinCarp Zorp seed round (aggregator)

Umbra umbraprivacy.com and docs; DefiLlama market data (15 Oct 2026)

Crucible Labs URnetwork monthly users, self-reported (21 Aug 2026)

The Daily Hodl Zach Pandl, Grayscale (2 Sep 2026)

UR Foundation SN25 subnet repository on GitHub